Q.No.	Sub. Sec.	Question Description
1		A self-driving car uses the image gradient to detect lane markings on the road. Explain how the Gradient magnitude $\|\boldsymbol{G}\|$ and orientation $\boldsymbol{\theta}$ help the system: 1. Find the edges of the lane lines, and 2. Decide the direction of those lines.
2		An autonomous vehicle navigation system using SIFT keypoints experiences tracking failures due to motion blur and illumination changes. Analyse how Orientation Assignment achieves rotation invariance, and explain how the Localization & Filtering stage uses Hessian matrices to eliminate unstable key point candidates along high-contrast edges.
3		Perform multi-level image segmentation on the following 4×4 grayscale image matrix I: $$I = \begin{bmatrix} 15 & 120 & 130 & 240 \\ 20 & 115 & 140 & 235 \\ 10 & 125 & 135 & 245 \\ 25 & 110 & 145 & 250 \end{bmatrix}$$ Using thresholds $T_0 = 50$ and $T_1 = 180$, segment the image using the rule: $$g(x,y) = \begin{cases} 0 & \text{if } I(x,y) \le 50 \quad \text{(Background)} \\ 128 & \text{if } 50 < I(x,y) \le 180 \quad \text{(Mid-tone)} \\ 255 & \text{if } I(x,y) > 180 \quad \text{(Foreground)} \end{cases}$$ **Tasks:** 1. Write down the resulting segmented matrix $g(x,y)$. 2. Calculate the pixel count for each segment class.
4		A healthcare organization has collected a large dataset of patient records to develop an intelligent disease prediction system; however, only a small number of records are labelled with confirmed disease outcomes while most records remain unlabelled—analyze the situation to determine the most appropriate learning approach among Supervised, Unsupervised, and Semi-supervised learning, and justify which of the Bayes, KNN, or ANN classifiers would be most suitable for developing an accurate and practical disease prediction system.
5		A computer-aided diagnosis system processes noisy, multi-scale CT or optical coherence tomography (OCT) volumes containing fine tubular structures (vessels, nerves) and textured tissue interfaces that vary widely in size and orientation. Propose and justify a hybrid multi-scale, multi-orientation feature extraction pipeline that combines: • Laplacian-of-Gaussian (LoG) / Difference-of-Gaussian (DoG) scale-space extrema • Steerable pyramid or oriented Gaussian derivative filter banks • Contourlet or Curvelet transform (or undecimated dual-tree complex wavelet transform) • At least one of ORB, BRISK, or KAZE (or dense SIFT/SURF descriptors) For each component, explain its specific contribution to scale and orientation selectivity. Critically compare the hybrid design against classical single-method baselines (Canny edge detection, Harris/Shi-

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		Tomasi corners, or pure Hough/Radon line detection) with respect to noise robustness, scale/rotation invariance, and feature discriminability. Provide a high-level algorithmic outline of the end-to-end pipeline and discuss its principal limitations and computational trade-offs.
6		Design a complete interactive segmentation pipeline for isolating a target anatomical structure from noisy multi-modal (CT/MRI/ultrasound) volumes that exhibit faint boundaries, intensity inhomogeneities, and partial-volume effects. Your pipeline must incorporate seeded region growing (or its variants), an active-contour / level-set edge-based method, and a graph-based energy minimization approach (Graph-Cut or Random Walker). Explain the distinct role of each technique, the recommended order of application, and precisely how user-supplied seeds or scribbles are injected and propagated. Justify the hybrid design in terms of leakage prevention across weak edges, suppression of over-segmentation, and generation of a topologically coherent binary mask that is directly usable by downstream detection or quantification modules. Include a high-level algorithmic outline and discuss the principal failure modes and computational considerations.

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